Electromagnetic Waves Introductory class

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Syllabus

- Electrostatics
- Magnetostatics
- Uniform plane waves
- Transmission line and waveguides

Textbooks

- Jr. Hayt, Buck, “Engineering Electromagnetics” 7th Edition, McGraw Hill
- Sadiku, “Elements of Electromagnetics”, 3rd Edition, Oxford University
- “Electromagnetics”, Schaum’s outlines

Instructions

- Maintain a separate notebook
- Bring calculator to every class
- Solve the homework problems every time
- Refer textbooks along with the PPTs
- Refer other section notes for more solved examples

Vector analysis and Coordinate systems

Scalars and Vectors

- Scalar refers to a quantity whose value may be represented by a single (positive or negative) real number

- Some examples include distance, temperature, mass, density, pressure, volume and time.

- A vector quantity has both a magnitude and a direction in space
- We especially concerned with two-and three-dimensional spaces only
- Displacement, velocity, acceleration and force are examples of vectors

- Scalar notation: A or A (*italic* or plain)
- Vector notation: $\mathbf{A}$ or $\vec{A}$ (**bold** or plain with arrow)

Vector Algebra



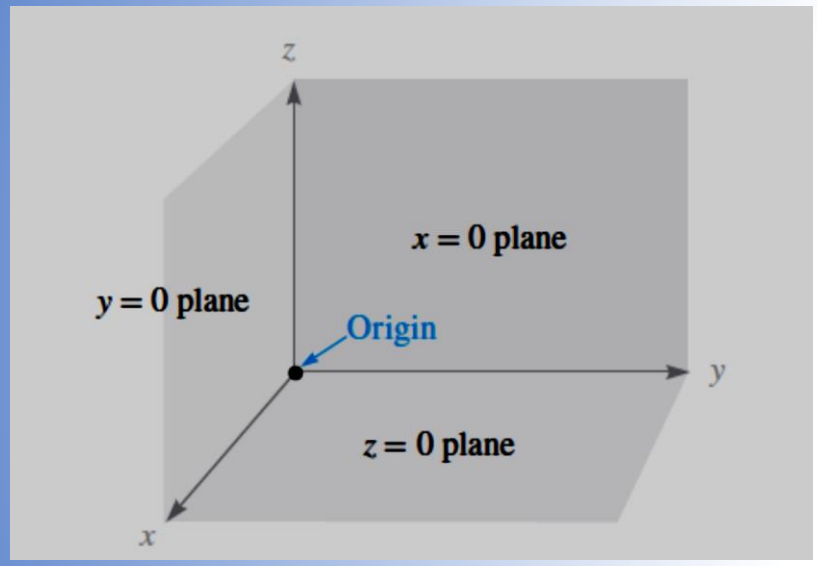
$$\mathbf{A}+\mathbf{B}=\mathbf{B}+\mathbf{A}$$

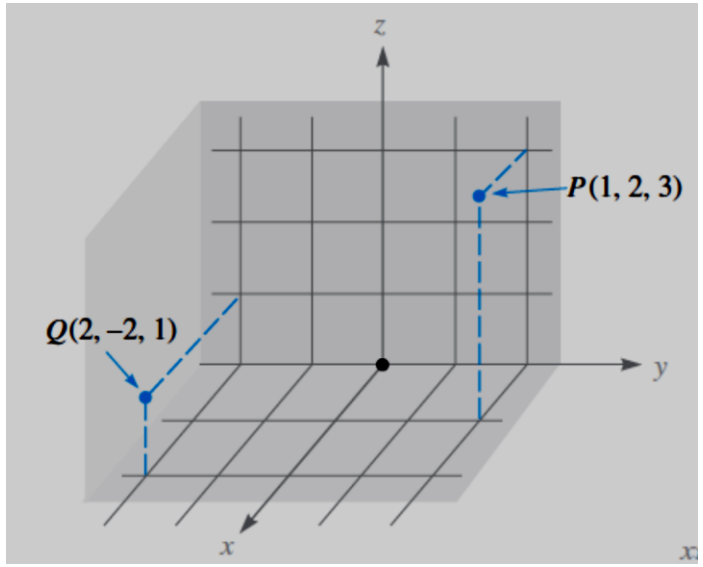
$$\mathbf{A}+(\mathbf{B}+\mathbf{C})=(\mathbf{A}+\mathbf{B})+\mathbf{C}$$

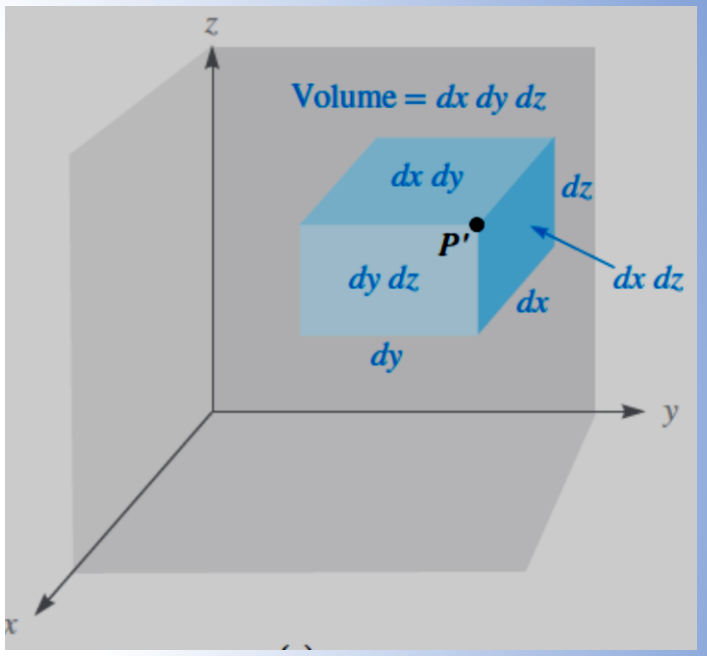
$$\mathbf{A}-\mathbf{B}=\mathbf{A}+(-\mathbf{B})$$

$$(r+s)(\mathbf{A}+\mathbf{B})=r(\mathbf{A}+\mathbf{B})+s(\mathbf{A}+\mathbf{B})=r\mathbf{A}+r\mathbf{B}+s\mathbf{A}+s\mathbf{B}$$

THE RECTANGULAR COORDINATE SYSTEM







- **Differential surface units:**
  $dx \cdot dy$
  $dy \cdot dz$
  $dx \cdot dz$
- **Differential volume unit :**
  $dx \cdot dy \cdot dz$

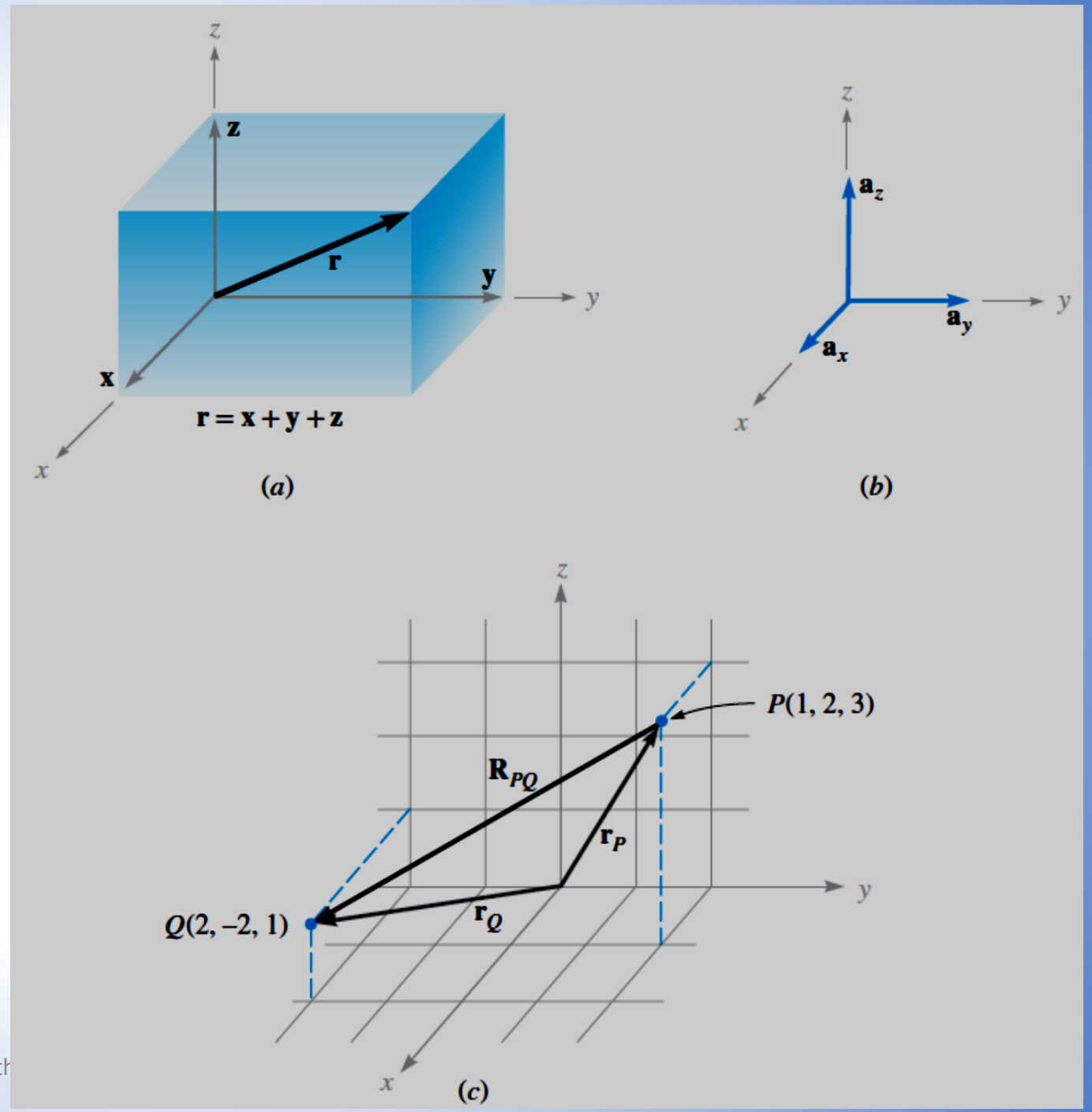
Vector components and unit vectors

$$\mathbf{r} = \mathbf{x} + \mathbf{y} + \mathbf{z}$$

$$\mathbf{r} = x\mathbf{a}_x + y\mathbf{a}_y + z\mathbf{a}_z$$

$\mathbf{a}_x, \mathbf{a}_y, \mathbf{a}_z$: unit vectors

$$\mathbf{R}_{PQ} = \mathbf{r}_Q - \mathbf{r}_P$$
$$= (2\mathbf{a}_x - 2\mathbf{a}_y + \mathbf{a}_z) - (1\mathbf{a}_x + 2\mathbf{a}_y + 3\mathbf{a}_z)$$
$$= \mathbf{a}_x - 4\mathbf{a}_y - 2\mathbf{a}_z$$



- For any vector **B**, $\mathbf{B} = B_x\mathbf{a}_x + B_y\mathbf{a}_y + B_z\mathbf{a}_z$:

$$|\mathbf{B}| = \sqrt{B_x^2 + B_y^2 + B_z^2} = B$$

Magnitude of **B**

$$\mathbf{a}_B = \frac{\mathbf{B}}{\sqrt{B_x^2 + B_y^2 + B_z^2}} = \frac{\mathbf{B}}{|\mathbf{B}|}$$

Unit vector in the direction of **B**

- Example

Given points $M(-1,2,1)$ and $N(3,-3,0)$, find $\mathbf{R}_{MN}$ and $\mathbf{a}_{MN}$.

$$\mathbf{R}_{MN} = (3\mathbf{a}_x - 3\mathbf{a}_y + 0\mathbf{a}_z) - (-1\mathbf{a}_x + 2\mathbf{a}_y + 1\mathbf{a}_z) = \underline{\underline{4\mathbf{a}_x - 5\mathbf{a}_y - \mathbf{a}_z}}$$

$$\mathbf{a}_{MN} = \frac{\mathbf{R}_{MN}}{|\mathbf{R}_{MN}|} = \frac{4\mathbf{a}_x - 5\mathbf{a}_y - 1\mathbf{a}_z}{\sqrt{4^2 + (-5)^2 + (-1)^2}} = \underline{\underline{0.617\mathbf{a}_x - 0.772\mathbf{a}_y - 0.154\mathbf{a}_z}}$$


Thank
you